

BHAVYA PATEL

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Professional Summary

- Graduate student in Applied Computer Science at Concordia University, specializing in Artificial Intelligence, Machine Learning, and Data Science.
- Over 2 years of professional experience in AI/ML engineering, data-driven development, and model deployment across academic and industry projects.
- Strong proficiency in Python, TensorFlow, Scikit-learn, and LangChain, with practical expertise in LLMs, NLP, and transformer-based architectures.
- Demonstrated leadership and teaching ability as a Teaching Assistant, coupled with strong communication and cross-functional collaboration skills.

Education

Concordia University

Master of Applied Computer Science (MApCompSc) – GPA: 3.33/4.0

Sept 2024 – Aug 2026

Montreal, QC, Canada

Charotar University of Science and Technology (CHARUSAT)

Bachelor of Technology in Computer Engineering – CGPA: 9.13/10

Jun 2020 – May 2024

Changa, India

Skills

- **Technical Skills:** Python, C, C++, Java, Shell Scripting, SQL, Django, Flutter, REST API Integration, UI/UX Design, Web Scraping, Data Analytics
- **AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, Keras, Hugging Face, Natural Language Processing (NLP), Vector Databases, Prompt Engineering, LLM Frameworks
- **Cloud & Tools:** AWS Bedrock, AWS SageMaker, Azure ML, Linux, Git, GitHub, Jira, Docker

Work Experience

Teaching Assistant

Concordia University

Sep 2025 – Present

Montreal, Canada

- Guided 20+ students in mastering Distributed Systems concepts such as Java RMI, CORBA, and SOAP-based Web Services through engaging lab demonstrations and personalized support.
- Facilitated weekly tutorials and 5+ lab sessions, aligning course content with practical exercises to maximize student understanding and participation.
- Evaluated 5+ assignments and final projects each term, ensuring consistent grading standards and delivering feedback within 5 days to foster continuous improvement.

Data Science Intern

Blue Data Consulting & IT Services Pvt. Ltd.

Dec 2023 – Jun 2024

Surat, India

- Devised an AI-driven workflow converting 50+ PowerPoint Presentations into video courses using LLMs, RAG, and Streamlit, increasing automation efficiency by 40%.
- Benchmarked 4 LLMs (GPT-3.5, Opus, Sonnet, Gemini), achieving 25% higher script accuracy and 35% faster generation with GPT-3.5-Turbo.
- Produced 10+ AI-based course modules (LangChain, Flowise AI, Synapse CoR, Prompt Engineering) in collaboration with a 5-member team, boosting content scalability by 3×.

Machine Learning Research Student

CHARUSAT University

Mar 2022 – Jun 2023

Changa, India

- Extracted and prepared 13K+ video frames from the UCF-101 Sports Action Dataset to train deep learning models for multi-sport activity recognition.
- Optimized the Vision Transformer (ViT) model, boosting classification accuracy from 84% to 94% through advanced hyperparameter tuning and data preprocessing.
- Investigated model performance across varying hyperparameter configurations and dataset augmentations, enhancing training stability and reducing validation loss by 9%.

Flutter Development Trainee

May – Oct 2022

Inspire Cyber Security

Surat, India

- Prototyped 3+ mobile applications in Adobe XD, converting wireframes into responsive Flutter interfaces that accelerated iteration cycles by 50%.
- Constructed a Notes App using SQLite for offline storage, implementing CRUD functionality with zero data loss during local syncing.
- Delivered a client project – "Palm Box Cricket Booking System" – enabling real-time slot reservations and increasing user engagement by 40% through an intuitive Flutter UI/UX.

Major Projects

Multilingual Video Script Generation Web App

Dec 2023 – May 2024

Python, LLMs, TTS APIs, Streamlit

- Formulated an AI pipeline that transformed PowerPoint slides into narrated videos using GPT models and TTS APIs.
- Integrated automated script extraction, multi-language selection (5+ languages), and animation into a single seamless workflow.
- Accelerated video production by 70%, embedding generated scripts into slide notes and exporting finalized outputs as MP4.

Museum View Detection - AI/ML

Dec 2024 – Mar 2025

PyTorch, CNN, XGBoost

- Constructed baseline classifiers (Decision Tree, Random Forest, XGBoost) to distinguish indoor vs. outdoor museum imagery.
- Developed and fine-tuned a PyTorch CNN on the MIT Places dataset (10,000 images) using advanced data augmentation and hyperparameter optimization.
- Attained 91.5% accuracy and 91.4% F1-score, surpassing baseline models by 16–22% in classification performance.

Publications

Vision Transformer in Sport Action: Recognizing Athletic Activities

Jun 2024

Springer Book Series – ICDSA 2024

- Co-authored a research paper introducing a novel Vision Transformer (ViT)-based approach for sports action recognition, enhancing classification accuracy across multiple athletic categories.
- Validated the model on a dataset of 10,000+ frames, achieving substantial performance improvements over traditional CNN architectures.
- Presented findings at the International Conference on Data Science Applications (ICDSA 2024), India, and published in the Springer Book Series. DOI: [10.1007/978-981-96-2179-8_35](https://doi.org/10.1007/978-981-96-2179-8_35)

Co-Curricular Activities

Professional Service Director

Sep 2022 – Aug 2023

Rotaract Club of CHARUSAT

- Coordinated and led 10+ social, technical, and cultural events, impacting over 500 students and community members.
- Directed team operations and event logistics for 10+ initiatives, improving coordination efficiency by 25% and strengthening leadership and conflict-resolution outcomes.

Certifications

- **Fundamentals of Digital Image and Video Processing** – Coursera – Northwestern University: Learned core concepts in signal processing, motion estimation, and video compression using MATLAB and Python.
- **Architecting Smart IoT Devices** – Coursera – EIT Digital: Gained practical knowledge in IoT system design, including sensor integration, edge computing, and secure communication protocols.

Additional project details, certifications, and GitHub links are available at:

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